

Teacher Guide — senior / module-13- indic-ecology

IKS Curriculum

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Overview

Indic Ecology — Vasudhaiva Kuṭumbakam

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Vasudhaiva Kuṭumbakam

Overview

Vasudhaiva Kuṭumbakam — “the world is one family” — from the Mahā Upaniṣad (6.71–73) anchors the Indian ecological tradition. Combined with the Pṛthivī Sūkta (Atharva Veda 12.1), Iṣopaniṣad mantra 1, and Bhagavad Gītā 3.10–14 on *yajña* as ecological cycle, students encounter classical frameworks and apply them to modern climate and sustainability challenges.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (local-ecology mapping)
- §11.8 — Holistic, multidisciplinary (ethics + ecology + climate)

Key terms (this module)

- *Vasudhaiva Kuṭumbakam* — “the world is one family” — Mahā Upaniṣad 6.71
- *yajña* — cycle of giving and receiving (BG 3.10–14)
- *ahiṃsā* — non-violence; extended to non-human life
- *devarakāṇḍu* — sacred grove (Karnataka, S. India)
- *sthāvara-jaṅgama* — immobile (plants, rocks) and mobile (animals) life

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper

- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — Vasudhaiva Kuṭumbakam: Identify the source verse and explain the principle.
- **Day 2** — Five Elements as Ecological Frame: Link Module 2 pañcabhūta to ecology — interconnection.
- **Day 3** — Rivers as Sacred: Discuss the ecological function of river-worship traditions.
- **Day 4** — Sacred Groves: Identify devarakāṣṭha and similar groves as biodiversity reserves.
- **Day 5** — Yajña — The Ecological Cycle: Read BG 3.10–14; map yajña to natural-cycle reciprocity.
- **Day 6** — Pṛthivī Sūkta (AV 12.1): Read the Earth-mother hymn and discuss imagery.
- **Day 7** — Iṣopaniṣad — Non-Greed Ethic: Translate *īśāvāsyam idaṁ sarvaṁ* and discuss its ecological implication.
- **Day 8** — Climate Change — The Crisis: Summarise key IPCC findings; what is changing and why.
- **Day 9** — From Ancient to Action: Design one local environmental project applying an Indic-ecology principle.
- **Day 10** — Project Pitch and Assessment: Present project pitches; summative quiz.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

10-day arc, 45 min per session.

Day 1 — Vasudhaiva Kuṭumbakam

Objective: Identify the source verse and explain the principle.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Five Elements as Ecological Frame

Objective: Link Module 2 pañcabhūta to ecology — interconnection.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Rivers as Sacred

Objective: Discuss the ecological function of river-worship traditions.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Sacred Groves

Objective: Identify devarakāṣṭha and similar groves as biodiversity reserves.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Yajña — The Ecological Cycle

Objective: Read BG 3.10–14; map yajña to natural-cycle reciprocity.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Pṛthivī Sūkta (AV 12.1)

Objective: Read the Earth-mother hymn and discuss imagery.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Iṣopaniṣad — Non-Greed Ethic

Objective: Translate *īśāvāsyam idaṁ sarvaṁ* and discuss its ecological implication.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Climate Change — The Crisis

Objective: Summarise key IPCC findings; what is changing and why.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — From Ancient to Action

Objective: Design one local environmental project applying an Indic-ecology principle.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Project Pitch and Assessment

Objective: Present project pitches; summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition’s framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji’s mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn’t?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Primary and secondary sources

- Mahā Upaniṣad 6.71–73 — vasudhaiva kuṭumbakam
- Atharva Veda 12.1 — Pṛthivī Sūkta
- Bhagavad Gītā 3.10–14 — yajña as cycle
- Iṣopaniṣad mantra 1 — *īśāvāsyam idaṁ sarvaṁ*
- Madhav Gadgil & Ramachandra Guha, *This Fissured Land: An Ecological History of India* (1992)
- IPCC Sixth Assessment Report (2021–2023)

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — Vasudhaiva Kuṭumbakam

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the source verse and explain the principle.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Vasudhaiva Kuṭumbakam* (IAST: Vasudhaiva Kuṭumbakam; lit. “the world is one family” — Mahā Upaniṣad 6.71”)

Primary source (today's reading)

From *Mahā Upaniṣad* 6.71–73 — *vasudhaiva kuṭumbakam* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “What's one thing you remember about yesterday's lesson?”
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **The verse.** From *Mahā Upaniṣad* 6.71–73: > *ayaṁ nijaḥ paro veti gaṇanā laghu-cetasām* / > *udāra-caritānām tu vasudhaiva kuṭumbakam* //

Translation: “*This is mine, that is another’s’ — such reckoning is for the narrow-minded. For the great-hearted, the entire earth is one family.*”

2. **Etymology.** *Vasudhā* — earth (literally “wealth-holder”). *Eva* — indeed. *Kuṭumbakam* — diminutive of *kuṭumba* (family), so “little family” or “the dear family.”
3. **Source clarification.** Also widely cited as appearing in the *Pañcatantra* and in *Hitopadeśa*. The *Mahā Upaniṣad* attestation is most commonly cited in academic contexts.
4. **The ecological reading.** Not just a humanist statement — it’s a principle of moral inclusion that extends to all life, all of nature. Read with the modern climate crisis in view, it becomes a foundational text for Indian environmental ethics.
5. **Critical caveat.** Pre-modern India was NOT universally eco-friendly. Forest clearance, soil degradation, and over-hunting also occurred. The TEXT articulates an ideal; the PRACTICE varied. Distinguish ethical framework from historical record.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Vasudhaiva Kuṭumbakam.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Five Elements as Ecological Frame

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: link module 2 *pañcabhūta* to ecology — interconnection.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *yajña* (IAST: yajña; lit. “cycle of giving and receiving (BG 3.10–14)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Recall Module 2’s pañcabhūta.** Earth, water, fire, air, space — the five elements seen as both physical and pervasive.
2. **The interconnection principle.** No element exists alone. Earth needs water to support life. Water needs space to flow. Air needs warmth (fire) to circulate. Space contains all. This is the COSMOLOGICAL grounding for ecological ethics.
3. **Modern translation.** Ecosystems = elemental cycles (carbon, nitrogen, water cycles). The pañcabhūta framework essentially names: matter (earth), water cycle (ap), energy (tejas), atmosphere (vāyu), and the systemic interconnection (ākāśa).
4. **Source.** Bhagavad Gītā 13.5 lists *mahā-bhūtāni* — the gross elements — as one of the *kṣetra* (field) categories. The conceptual frame is in place by ~2nd century BCE.
5. **Discussion.** “*If we damage one element — pollute air, contaminate water, deforest land — what happens to the others?*” Modern climate science answers concretely. “*Indic ecology says these aren’t separate problems.*”

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Five Elements as Ecological Frame.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Rivers as Sacred

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: discuss the ecological function of river-worship traditions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ahimsā* (IAST: *ahimsā*; lit. “non-violence; extended to non-human life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Indian river tradition.** *Gaṅgā, Yamunā, Sarasvatī, Narmadā, Kāverī, Godāvarī, Sindhu* — seven sacred rivers in Hindu tradition. Each has festival cycles, pilgrimage routes, ritual practices.
2. **The ecological function of sacredness.** Treating a river as sacred → norms against pollution, against blocking the flow, against over-extraction. *In principle*. These norms are NOT always honoured today (Ganga, Yamuna are heavily polluted).
3. **Modern parallel.** Ecuador and Bolivia have laws giving rivers legal personhood. Aotearoa/New Zealand granted legal personhood to the Whanganui River in 2017.

India's Ganga and Yamuna were granted similar status briefly in 2017 (court ruling overturned). The legal innovation echoes the older sacred-river framing.

4. **Source.** Atharva Veda Pṛthivī Sūkta 12.1 — multiple references to flowing waters. Vedic hymns to *āpaḥ* (waters). Bhāgavata Purāṇa discusses the seven rivers.
5. **Critical look.** “*Why are sacred-river norms often violated in practice today? What changes in society have weakened the older protections? Can the framework be revived for the climate crisis?*”

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Rivers as Sacred.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Sacred Groves

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify *devarakāṣṭha* and similar groves as biodiversity reserves.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *devarakāḍu* (IAST: *devarakāḍu*; lit. “sacred grove (Karnataka, S. India)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define sacred grove.** Patches of native forest preserved by community tradition, often associated with a deity, sage, or ancestor. Hunting, cutting, and even firewood collection are prohibited.
2. **Indian examples.**
 - *Devarakāḍu* (Karnataka)
 - *Kāvu* (Kerala)
 - *Devarakad* (Maharashtra)
 - *Sarna* (Jharkhand, Odisha, Bihar — tribal areas)
3. **Ecological function.** Despite covering <0.1% of India’s forest area, sacred groves preserve disproportionate biodiversity: endemic species, medicinal plants, water-table recharge zones. Madhav Gadgil’s research (1992 and onward) documented this.
4. **Source.** *Pr̥thivī Sūkta* (Atharva Veda 12.1) — particularly verses 11–13 on forests. Modern source: Gadgil & Guha, *This Fissured Land* (1992).
5. **Activity.** Each student researches one sacred grove (or biodiversity site) in their region and prepares a 1-page report. Map the grove, list flora/fauna, document traditional protection rules.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see **activities/**). Today’s slice: focus on the part of the activity that reinforces Sacred Groves.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Yajña — The Ecological Cycle

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: read bg 3.10–14; map yajña to natural-cycle reciprocity.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sthāvara-jaṅgama* (IAST: sthāvara-jaṅgama; lit. “immobile (plants, rocks) and mobile (animals) life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The Gītā passage.** Bhagavad Gītā 3.10–14. Read the key verses: > *annād bhavanti bhūtāni parjanyaś anna-sambhavaḥ* / > *yajñād bhavati parjanya yajñaḥ karma-samudbhavaḥ* // > “From food beings come, from rain food comes, from yajña rain comes, yajña comes from action.” (3.14)
2. **Decode the cycle.** Yajña → rain → food → beings → action (work) → yajña (giving back). A closed loop where each takes AND gives. This is the ecological cycle in Sanskrit terms.

3. **Modern parallel.** Carbon cycle: plants take CO₂ → animals eat plants → animals exhale CO₂ → plants take CO₂. Water cycle: evaporation → condensation → rain → soil → evaporation. The Gītā verse abstracts this into a yajña frame.
4. **Source check.** Multiple Upaniṣads also discuss the give-and-receive principle. Cite verifiable references in `sources.md`.
5. **Modern relevance.** Climate ethics. “If we take from nature without giving back, the yajña is broken. What does ‘giving back’ look like in 2026?” (Plant trees, reduce waste, restore water bodies, etc.)

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Yajña — The Ecological Cycle.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 6 — Pṛthivī Sūkta (AV 12.1)

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: read the earth-mother hymn and discuss imagery.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Vasudhaiva Kuṭumbakam* (IAST: Vasudhaiva Kuṭumbakam; lit. “the world is one family” — Mahā Upaniṣad 6.71”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The text.** Atharva Veda 12.1 — *Ṛṥhivī Sūkta* (Hymn to the Earth). 63 verses. Most extensive ancient ecological text in the Vedic corpus.
2. **Key verses.** Read aloud (in IAST + English): > *mātā bhūmiḥ putro ahaṃ ṛṥhivyāḥ* — “Earth is my mother, I am her son.” (12.1.12) > *yat te bhūme vikhanāmi kṣipraṃ tad api rohatu* — “What I dig from the Earth, may it grow back swiftly.” (12.1.35)
3. **The framing.** Earth as mother. Resource extraction as legitimate but bounded by replenishment. Cleanliness, beauty, biodiversity as goods to preserve.
4. **Cross-cultural.** Earth-mother concept appears in many cultures: Pachamama (Andean), Gaia (Greek). The Atharva Veda articulation is older than most extant references.
5. **Critical reading.** The hymn is a religious text. It articulates an ETHIC, not an empirical claim. “*Does treating Earth as mother make us better stewards? Or does it sometimes lead to passive acceptance of destructive practices (‘Earth will heal’)?*” Both views exist.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Ṛṥhivī Sūkta* (AV 12.1).

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Iṣopaniṣad — Non-Greed Ethic

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: translate *īśāvāsyam idaṁ sarvaṁ* and discuss its ecological implication.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *yajña* (IAST: yajña; lit. “cycle of giving and receiving (BG 3.10–14)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The text.** *Īśāvāsyā Upaniṣad* (a short Upaniṣad, 18 verses), mantra 1: > *īśāvāsyam idaṁ sarvaṁ yat kiñca jagatyāṁ jagat / > tena tyaktena bhuñjīthā mā gṛdhaḥ kasya svid dhanam //*

Translation: “*All this — whatever moves in this moving world — is enveloped by the Lord. Enjoy through renunciation. Do not covet anyone’s wealth.*”

2. **The ecological reading.** *Tyaktena bhuñjīthā* — “enjoy by renouncing.” Use the world, but with restraint. Take only what you need; leave the rest. This is one of the strongest non-greed ethics in any wisdom tradition.
3. **Modern parallel.** Steady-state economics (Herman Daly). Voluntary simplicity. Donut economics (Kate Raworth). The principle: don’t grow beyond ecological limits.

4. **Source.** Isāvāsya Upaniṣad is part of the Yajurveda. Translations: Eknath Easwaran, Stephen Phillips academic editions.
5. **Discussion.** “How does ‘enjoy by renouncing’ apply to your own life? Phone usage? Food? Clothes? Travel?” Each student writes 3 sentences. NO right answer — the point is HONEST self-reflection.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Iṣopaniṣad — Non-Greed Ethic.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Climate Change — The Crisis

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: summarise key ipcc findings; what is changing and why.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *ahimsā* (IAST: ahimsā; lit. “non-violence; extended to non-human life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The science.** Project 3 key IPCC AR6 graphs:
 - Global temperature anomaly since 1850 (~+1.2°C and rising).
 - CO2 atmospheric concentration since 1900 (290 → 420+ ppm).
 - Sea-level rise (~20 cm since 1900).
2. **The agreement.** 99%+ of climate scientists agree: warming is human-caused, primarily fossil-fuel emissions. The IPCC reports synthesise thousands of studies.
3. **What’s at stake for India.** Higher monsoon variability, glacier retreat in the Himalayas, sea-level rise threatening coastal cities, heat-stress in agriculture. NDCs (Nationally Determined Contributions) target net-zero by 2070.
4. **Source.** IPCC Sixth Assessment Report (Working Groups I, II, III; 2021–2023). NCERT Geography XI ch. 11.
5. **Bridge to Indic frameworks.** “*What does vasudhaiva kuṭumbakam say about climate change? What does yajña-cycle suggest about our daily responses? What does īsopaniṣad teach about consumption?*” Each student writes one paragraph connecting one ancient text to one modern climate response.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see **activities/**). Today’s slice: focus on the part of the activity that reinforces Climate Change — The Crisis.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — From Ancient to Action

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: design one local environmental project applying an indic-ecology principle.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *devarakāḍu* (IAST: devarakāḍu; lit. “sacred grove (Karnataka, S. India)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Project design phase.** Students apply Module 13’s principles to design ONE local environmental project:
 - Plant-a-tree drive (yajña reciprocity)
 - Sacred-grove documentation (biodiversity preservation)
 - Water audit at school / home (resource consciousness)
 - Waste-sorting station (extending ahimsā to materials)
 - Local-language climate-awareness poster (cultural communication)
2. **Project criteria.** Must (a) be achievable in 4 weeks, (b) cite at least ONE classical Indic principle, (c) document baseline and outcome data, (d) include a community/family component.
3. **Pair-brainstorm.** Pairs spend 10 minutes generating project ideas, then sketch one.

4. **Source-anchoring.** Each project must reference a specific verse, hymn, or principle from Modules 13's source list.
5. **Tomorrow's pitch.** Each student presents their project pitch in 3 minutes. Refine tonight.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces From Ancient to Action.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Project Pitch and Assessment

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present project pitches; summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sthāvara-jāṅgama* (IAST: *sthāvara-jāṅgama*; lit. “immobile (plants, rocks) and mobile (animals) life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Project pitches (3 min each).** Each student presents their environmental-action project: problem identified, principle anchored, plan, expected outcome, measurement.
2. **Q&A and feedback (1 min each).** Peers ask one question and offer one suggestion.
3. **Voting.** Class votes on 3 favourite projects. These will be amplified (school-wide, parent-day presentations).
4. **Summative quiz (20 min).** See quizzes/summative.md.
5. **Module close.** “*Of all the texts and ideas in this module, which one will stay with you the longest? Why?*” Each student writes 2 sentences.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see activities/). Today’s slice: focus on the part of the activity that reinforces Project Pitch and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Formative Quiz — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Indic Ecology — Vasudhaiva Kuṭumbakam in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Rivers as Sacred’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Indic Ecology — Vasudhaiva Kuṭumbakam is purely a religious concept and has no scientific framing.*
5. Connect Indic Ecology — Vasudhaiva Kuṭumbakam to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘Vasudhaiva Kuṭumbakam’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Indic Ecology — Vasudhaiva Kuṭumbakam and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *Vasudhaiva Kuṭumbakam* in your own words. A2. Define *yajña* in your own words. A3. Define *ahimsā* in your own words. A4. Define *devarakāṇḍu* in your own words. A5. Define *sthāvara-jāṅgama* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Indic Ecology — Vasudhaiva Kuṭumbakam in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain

why it matters. B4. Compare Indic Ecology — Vasudhaiva Kuṭumbakam with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Indic Ecology — Vasudhaiva Kuṭumbakam from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Vasudhaiva Kuṭumbakam** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Five Elements as Ecological Frame** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Rivers as Sacred** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Sacred Groves** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Yajña — The Ecological Cycle** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Pṛthivī Sūkta (AV 12.1)** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Iṣopaniṣad — Non-Greed Ethic** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Climate Change — The Crisis** · Write 2 sentences about today's concept. (~10 min)

9. **Day 9 — From Ancient to Action** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Project Pitch and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Indic Ecology — Vasudhaiva Kuṭumbakam to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Indic Ecology — Vasudhaiva Kuṭumbakam with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Indic Ecology — Vasudhaiva Kuṭumbakam? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Sacred Grove (or Local Ecology) Mapping

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Indic Ecology — Vasudhaiva Kuṭumbakam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

large paper, coloured markers, optional photos

What students do

Identify one sacred grove, river-bank, or biodiversity site in your region. Map it with notes on flora, fauna, traditional protection practices, and current threats.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Local Ecology Action Plan

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior

What you will do

Design and pitch one practical environmental project for your school or neighbourhood (e.g. plant-a-tree drive, water-audit, waste-sorting station). Pitch must reference one Indic principle (vasudhaiva kuṭumbakam, yajña, sacred-grove model).

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.